

Block diagram

sheet symbols open the pages; arrows are the interfaces between them

Power and ESP32-S3 DevKit

ESP32-S3 DevKitC-1 (N16R8): supply pins, reserved GPIO
5V0 -> FB1 -> LP5907 -> 3V3_RADIO -> VMCU_IN
DevKit 3V3 decoupling, test points

Fajl: power.kicad_sch

The ESP32 module is split in four units, each on the page it serves:
IC1A supply (p2), IC1B radio link (p3), IC1C user interface (p4), IC1D reserved (p2).

Radio board (BRD4265B)

P200/P201 to the BRD4265B (EFR32FG23)
FG23 SPI (IO_MUX slave) + UART + RDY
SWD bit-bang, I2C board-ID, cut points JP1-JP8

Fajl: radio.kicad_sch

User interface

ER-TFTM024-3 TFT, 4-wire SPI + XPT2046 touch
SSD1306 OLED on I2C 0x3C
two ADC1 button ladders (6 keys)

Fajl: ui.kicad_sch

Mechanical

M3 mounting holes, not plated to GND

H1
M3
H2
M3
H3
M3
H4
M3

Design notes

Net names are global (labels and rail symbols); layout, twin and firmware share one name per net.

A label sits on a short stub at a pin end; wires stay inside a block; supplies only as rail symbols.

GPIO plan: FG23 link on the SPI2 IO_MUX pins 10/11/12 + UART TX 13 + RDY 14; TFT on SPI3 via the
matrix (4-7, 15-18); touch 8/9/21; buttons ADC1 1/2; I2C 41/42; SWD 38/39/48, RESETn 40.

Forbidden: strapping 0/3/45/46, octal PSRAM 35-37 (N16R8). Reserved: UART0 43/44, USB 19/20, WS2812 47.

Cut points JP1-JP8 isolate every radio-board signal; R1-R5 are OR links on a damping-resistor footprint.

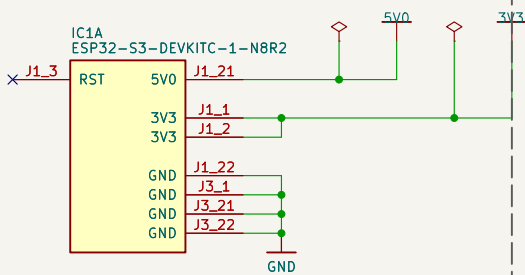
Assembly: R26/R27 DNP, value NA(4.7k). Values carry the critical parameter (value/voltage/tolerance).

FB1: value 600R@100MHz but MPN field BLM18SP300SH1D (30R) - resolve before ordering.

ESP32-S3 DevKitC-1

supply pins: the DevKit is the power source

RST = EN, button on the DevKit

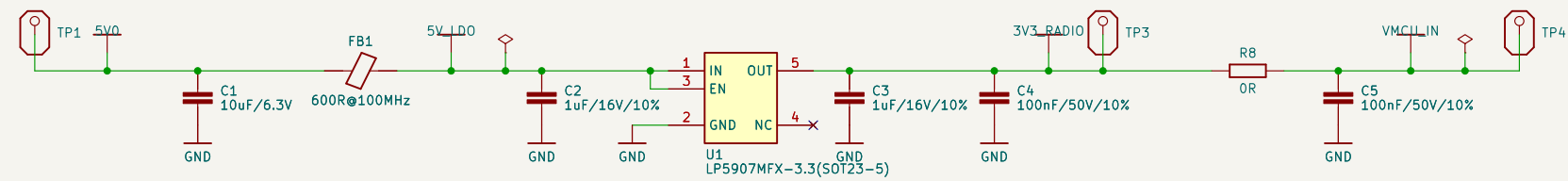


GND test point and ERC flag



Radio supply

5V0 -> FB1 -> LP5907 -> 3V3_RADIO -> R8 -> VMCU_IN



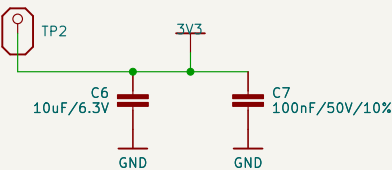
FB1 + C1/C2 keep DevKit 5 V noise out of the LDO

LP5907: 250 mA, 6.5 uVrms, EN tied to IN

R8 OR fitted: VMCU_IN = 3V3_RADIO (P201 pin 2)

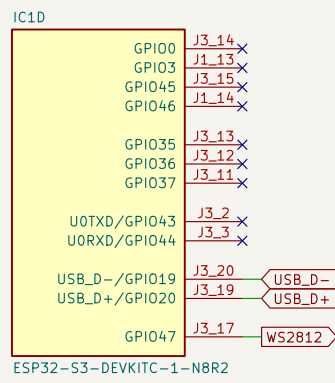
3V3 decoupling

DevKit LDO output: displays and pull-ups



Reserved GPIO

not used on this board - keep it that way



strapping pins (GPIO0/3/45/46): leave open

octal PSRAM of the N16R8 module: forbidden

UART0 = DevKit USB-UART console

native USB on the DevKit connector

RGB LED on the DevKit (AliExpress clone: GPIO47)

ESP32-S3 radio link

SPI2 IO_MUX slave, UART, SWD, I2C

ESP32-S3-DEVKITC-1-N8R2

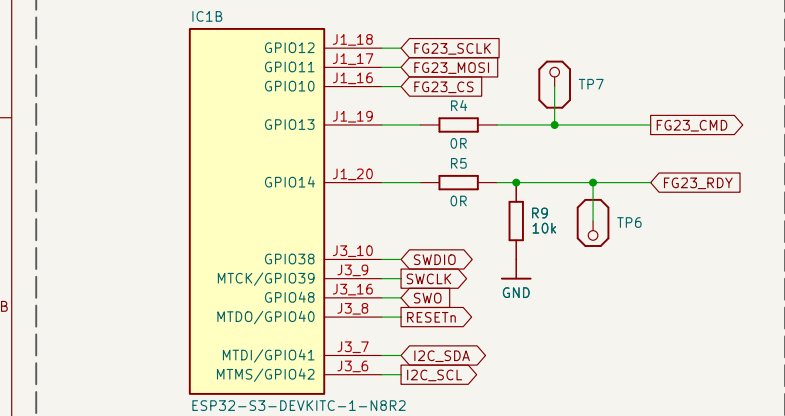
ESP32 is the SPI slave (FG23 = master),
IO_MUX pins only – no GPIO matrix.

R4/R5 near the ESP32; R9 near P201.

SWD is bit-banged; RESETn open-drain.

Laid out for the AliExpress DevKit clone (LED on GPIO47). Official DevKitC-1 puts the WS2812 on GPIO38 (v1.1) / GPIO48 (v1.0) = SWDIO / SWO here: remove that LED or use the clone.

SPI2 IO_MUX slave, UART, SWD, I2C



ESP32 is the SPI slave (FG23 = master),
IO_MUX pins only – no GPIO matrix.

R4/R5 near the ESP32; R9 near P201.

SWD is bit-banged; RESETn open-drain.

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P201 – radio board header

FG23 link, SWD, VMCU (BRD4265B expansion)

CON2
TFC-120-02-F-D-A-K-TR

1 2
3 4
5 6
7 8
9 10
11 12
13 14
15 16
17 18
19 20
21 22
23 24
25 26
27 28
29 30
31 32
33 34
35 36
37 38
39 40

GND

FG23_SPARE_RB

FG23_RDY

FG23_CMD

FG23_CS

FG23_SCLK

R3 OR

R2 OR

SWO_RB

SWDIO_RB

SWCLK_RB

R1

OR

FG23_MOSI

TP8

VRF_IN

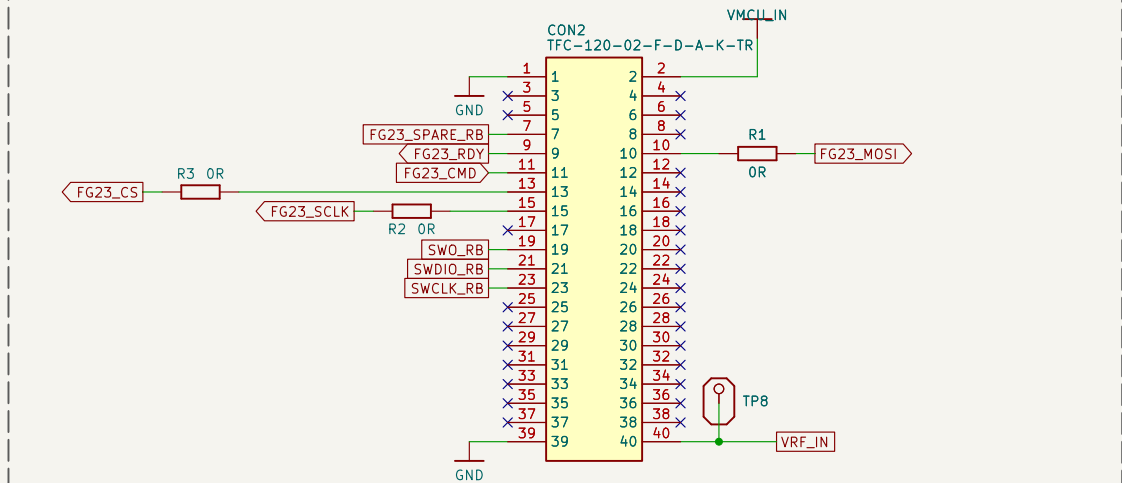
GND

R1-R3 sit at P201: 0R fitted,
22-68R damping footprint (2-layer build: 68R).

Pinout = Silabs BRD4265B / WSTK expansion header.

VRF_IN (P201.40): track to TP8 only – a test
point on the BRD4265B, not connected there.

FG23 link, SWD, VMCU (BRD4265B expansion)



R1-R3 sit at P201: 0R fitted,
22-68R damping footprint (2-layer build: 68R).

Pinout = Silabs BRD4265B / WSTK expansion header.

VRF_IN (P201.40): track to TP8 only - a test point on the BRD4265B, not connected there.

P200 – radio board header

supply, reset, I2C (board-ID EEPROM)

3V3_RADIO_RB

CON1
TFC-120-02-F-D-A-K-TR

RESETn_RB

5V_RADIO

I2C_SCL_RB

I2C_SDA_RB

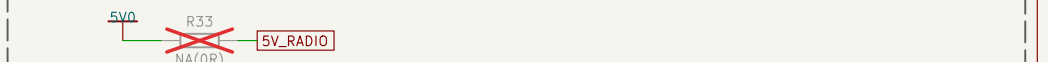
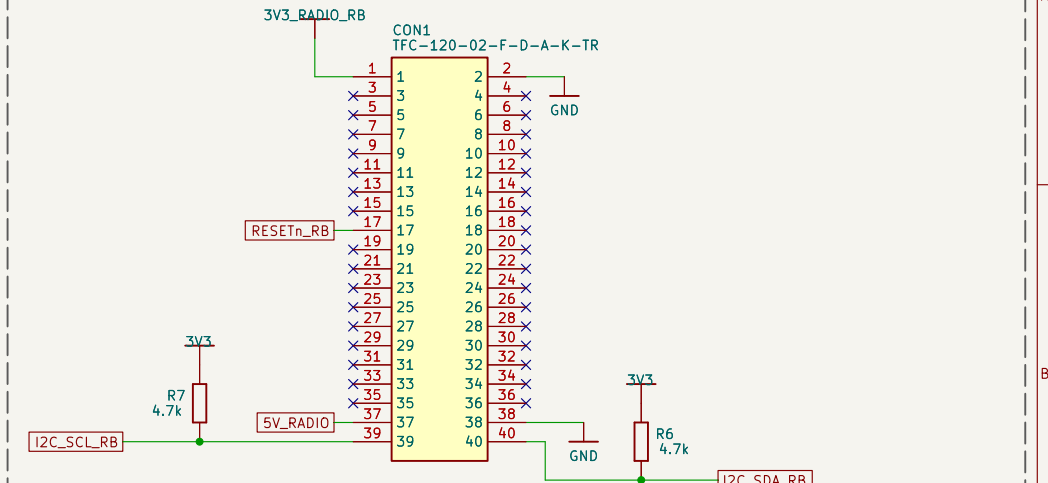
5V0

R33
NA(OR)

5V_RADIO

R6/R7: I2C pull-ups on the radio side.
Board-ID EEPROM on the radio board, OLE 0x3C on ours.

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supply, reset, I2C (board-ID EEPROM)
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R33 not fitted: DevKit 5 V to P200.37,
unused on the BRD4265B.

R6/R7: 12C pull-ups on the radio side.
Board-ID EEPROM on the radio board, OLED 0x3C on ours.

Cut points

one solder jumper per radio-board signal, bridged by default

3V3_RADIO

C8
100nF/50V/10%

GND

JP1
Bridged

3V3_RADIO_RB

RESETn

JP2
Bridged

RESETn_RB

JP3
Bridged

I2C_SCL

I2C_SCL_RB

JP4
Bridged

I2C_SDA

I2C_SDA_RB

JP5
Bridged

FG23_SPARE

FG23_SPARE_RB

JP6
Bridged

SWO

SWO_RB

JP7
Bridged

SWDIO

SWDIO_RB

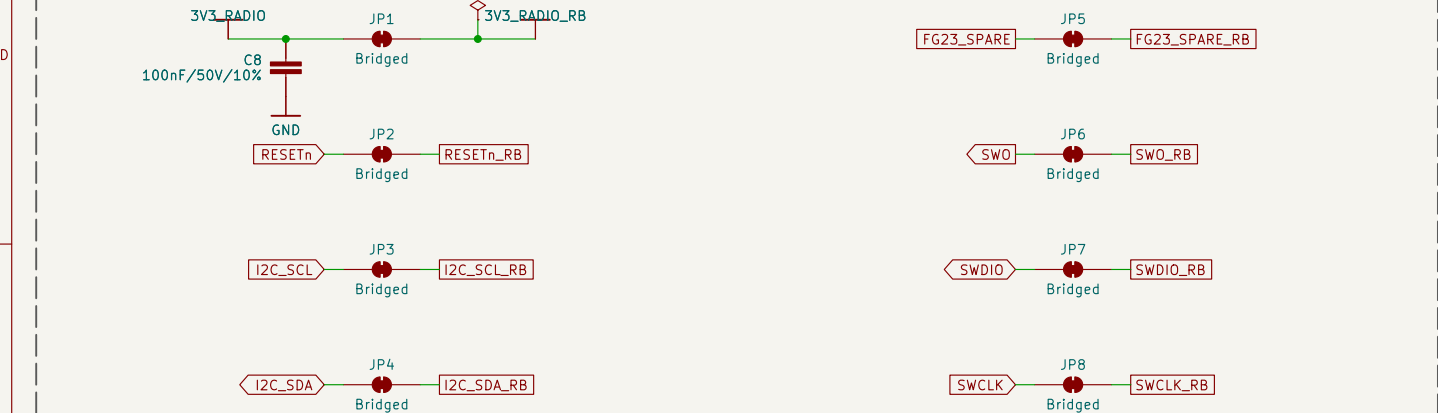
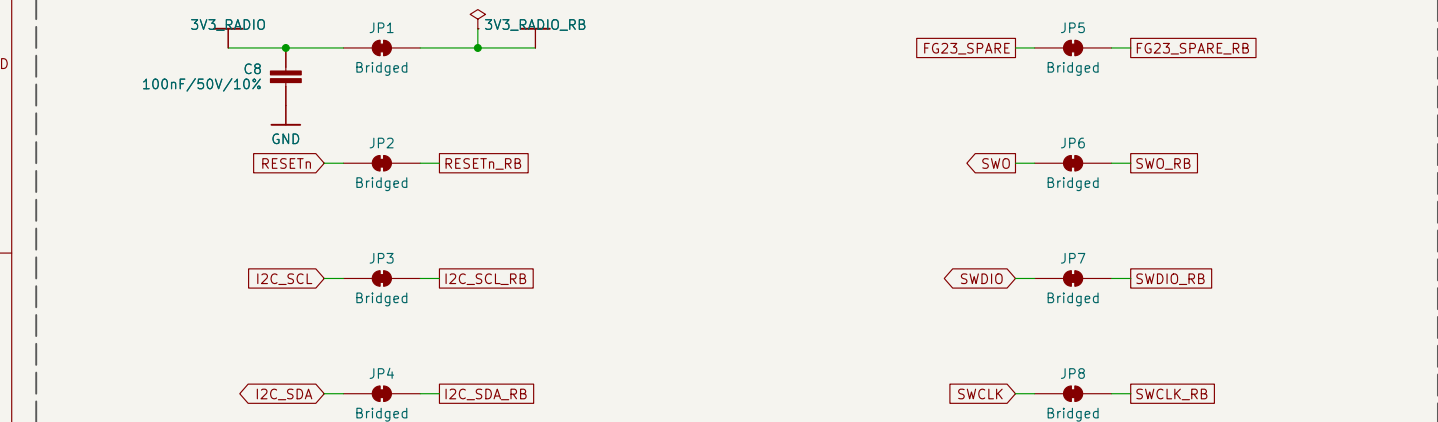
JP8
Bridged

SWCLK

SWCLK_RB

FG23_SPARE: track and test pad only, no part fitted.

one solder jumper per radio-board signal, bridged by default



FG23_SPARE: track and test pad only, no part fitted.

P200/P201 mezzanine, FG23 link, SWD, cut points

RF Sparkling Kft. – HA7DCD

Sheet: /Radio board (BRD4265B)/

Title: ionos-sdr-mini rev-B3 – Radio board (BRD4265B)

Size: A3	Date: 2026-10-09	Rev: B3
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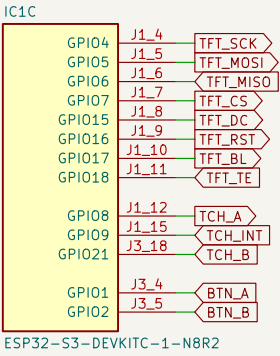
Size: A3	Date: 2026-10-09	Rev: B3
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Size: A3	Date: 2020-10-09	Rev: 03
KiCad E.D.A. 10.0.6		Id: 3/4

Size: A3	Date: 2020-10-09	Rev: 03
KiCad E.D.A. 10.0.6		Id: 3/4

ESP32–S3 user interface

TFT on SPI3 via the GPIO matrix

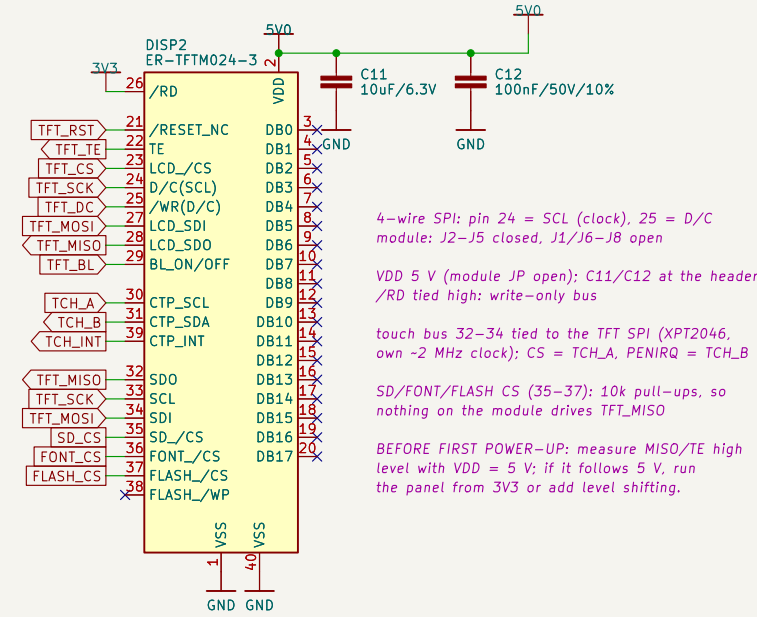


Touch, no jumper – firmware picks the mode:
TCH_A GPIO8 resistive CS / CTP SCL
TCH_B GPIO21 PENIRQ / CTP SDA
TCH_INT GPIO9 unused / CTP INT
R31/R32 4.7k fitted in every build.

BTN_A/BTN_B: ADC1, usable with WiFi on.

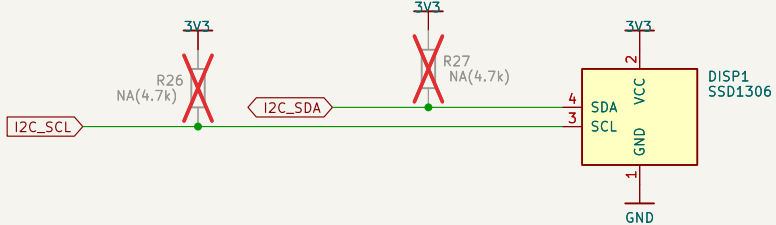
TFT ER–TFTM024–3

2.4 in 240x320, 4–wire SPI, resistive touch (XPT2046)



OLED SSD1306

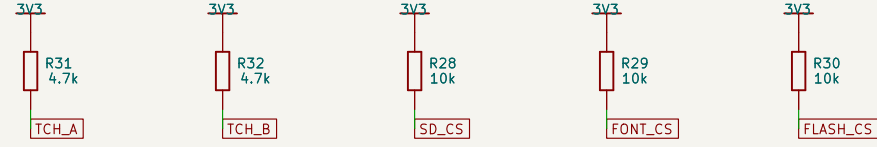
0.96 in 128x64, I2C 0x3C



R26/R27 not fitted: the bus pull–ups are R6/R7 on the radio side; fit them if JP3/JP4 are cut.

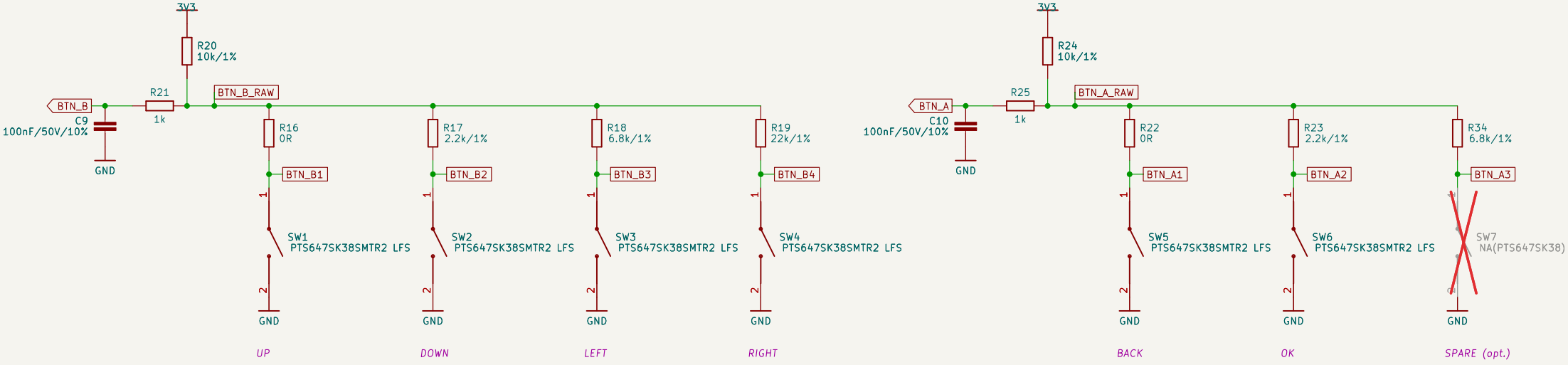
Header pull–ups

touch CS/IRQ and the module flash/font/SD chip selects



Buttons

two resistor ladders on ADC1: one GPIO per ladder



Both ladders: OR / 2.2k / 6.8k / 22k under a 10k pull–up → one firmware table:
levels 0 / 0.60 / 1.34 / 2.27 / 3.3 V, thresholds 0.30 / 0.97 / 1.80 / 2.79 V. Wake: OR keys only.

TFT + resistive touch, OLED, two ADC button ladders

RF Sparkling Kft. – HA7DCD

Sheet: /User interface/

File: ui.kicad_sch

Title: ionos–sdr–mini rev–B3 – User interface

Size: A3

Date: 2026–10–09

Rev: B3

KiCad E.D.A. 10.0.6

Id: 4/4